ADOBE UNIVERSITY HACKATHON 2026

Chameleon

Any design. Every canvas.

A layout engine for Adobe Express that moves the furniture instead of stretching the photo.

Resizing today stretches the design



The source: a 1080×1080 post



Naive stretch: squashed shapes, unreadable copy



Chameleon: the same pieces, placed again for the new shape

Resizing is a furniture problem: keep the pieces, place them again

What belongs together

Buttons, stickers on products, captions and cards, recovered with no labels

What matters most

Headline, logo and call to action outrank badges and fine print

How every line wraps

Measured with the renderer’s own fonts, so each line break is the one you’ll see

How the engine works

1 Analyse	Elements become units that travel together; importance comes from type, size and position
2 Order	An XY-cut recovers reading order, side-by-side rows and the column each unit aligns to
3 Lay out	Stack, split or banner row by target shape, at the largest scale that keeps every text legible
4 Degrade	When space runs out, the least important unit steps aside, and only if that helps the rest
5 Finish	Full-bleed crops, decorations anchored, images in proportion, everything inside the canvas

The hardest sample, eight formats

A 1080×1350 poster with a 150-character headline, three badges, a rotated ribbon and a missing image



Instagram 1080×1080



Story 1080×1920



Facebook 1200×630



Leaderboard 728×90



Med. rectangle 300×250



Half page 300×600



LinkedIn 1584×396



Email 600×200

It holds up everywhere

200

resizes in the test suite: 10 designs × 20 real formats, every hard rule asserted

1,500

randomly generated designs: no crash, no rule violation, no fallback

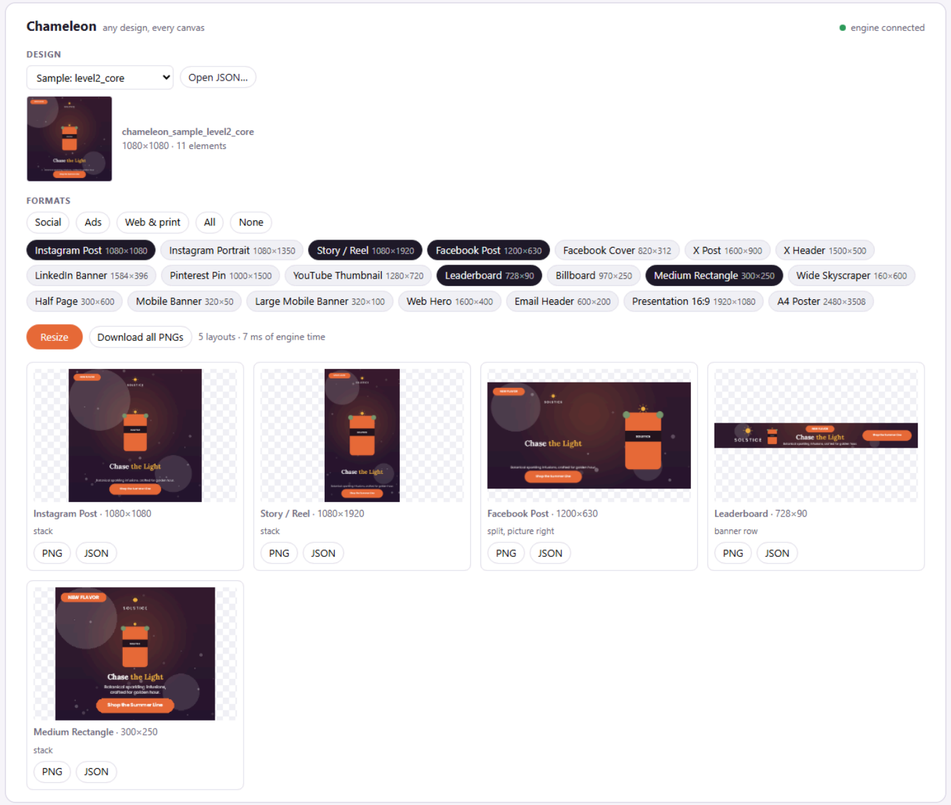
1–150 ms

per layout: deterministic, offline, no model

0

pixels generated: only geometry and type size change

In Adobe Express, on the page you're editing



The same panel in a browser, running the engine

One click, every format

Each chosen format becomes a real, editable page: text stays text, nothing is flattened

Nothing is lost

Elements set aside are hidden, not deleted, so one click brings them back

It explains itself

Every result says what it did: "banner row · 3 secondary elements set aside"

What comes next

Several options per format

The engine already scores competing layouts: show the best three and let the designer pick

Lock what matters

Pin a logo or headline in place and let the engine arrange everything else around it

See the focal point

A small vision model for smart photo crops and hints about each element’s role

Copy that fits

Shorter, approved variants of long copy for the tiniest ad slots

Engine, Express add-on, tests and demo video in one repository · Any design. Every canvas.